

Product Requirements Document (PRD)

Project Name: DoomOS (KDE Plasma Edition)

Document Version: 1.0.0

Status: Approved for Architecture & Design

Target Environments: Modern x86_64 Desktops, Laptops, and Handhelds

Author: Senior Systems Engineer / Technical Lead

1. Executive Overview & Mission Statement

DoomOS is a next-generation, performance-tuned Linux distribution engineered to bridge the gap between uncompromising power-user performance and an intuitive, foolproof out-of-the-box experience. Built on top of a cutting-edge **KDE Plasma 6 (Wayland-native)** desktop environment, DoomOS directly eliminates the systemic friction points that drive users away from Linux—specifically targeting:

- Gamers:** Eliminating driver breakage, Wayland frame pacing stutter, controller friction, and HDR hurdles.
- Everyday Users:** Eliminating update anxiety with automated snapshot rollbacks, blurry fractional scaling, battery drain, and codec headaches.
- Developers:** Providing seamless Wayland screen sharing with system audio, pre-configured rootless containers, and optimized kernel IPC/inotify limits.

2. Target Personas & Core Use Cases

TARGET USERS		
THE GAMER	THE EVERYDAY USER	THE DEVELOPER
High refresh, low latency	Rock-solid stability, battery	Fast toolchains, Docker,
Steam, Proton, G-Sync/HDR	Netflix, Office, App Store	Multi-monitor, Portals

Persona 1: The Modern Gamer ("Alex")

- Hardware:** Modern AMD/Intel CPU + NVIDIA RTX 3000/4000 or AMD Radeon RX 7000 series GPU, 144Hz+ VRR display.
- Goals:** Wants to install the OS, launch Steam, enable HDR, plug in an Xbox/PS5 controller, and play without editing config files or facing black screens after kernel updates.

Persona 2: The Daily Driver ("Taylor")

- Hardware:** Ultrabook / Laptop (Framework, Dell XPS, ThinkPad, ASUS Zenbook) with high-DPI display.

- **Goals:** Requires reliable suspend/resume without battery drain in backpacks, crystal-clear 125%/150% UI scaling, universal media playback, and an assurance that system updates will never brick their workstation.

Persona 3: The Full-Stack / Systems Developer ("Marcus")

- **Hardware:** Multi-monitor developer workstation or laptop, running Docker/Podman, VS Code, JetBrains IDEs, and heavy compilation workloads.
- **Goals:** Demands instant Wayland screen sharing with audio for remote meetings, high inotify file watch limits for large mono-repos, rootless container execution out-of-the-box, and isolated sandbox development environments.

3. System Architecture Specification

USERLAND INTERACTION		
KDE Plasma 6 (KWin Wayland)	Calamares Offline/Online Installer	KDE Discover / CLI
PLATFORM SUBSYSTEMS		
DoomOS Guardian (DKMS check)	Snapper / Btrfs Hooks	PipeWire Audio/Video Wire
Auto-CPUFreq / Power Profiles	xdg-desktop-portal-kde	Flatpak Sandbox Runtime
INIT & RUNTIME		
systemd (PID 1) D-Bus Message Bus Polkit ACL		
KERNEL & HARDWARE ABSTRACTION		
Linux Kernel (LTS / Zen low-latency)	Btrfs Filesystem	ZRAM Compressed Swap
Mesa Gallium/Vulkan	NVIDIA Open/DKMS Module	xone / DualSense HID Drivers

4. Non-Functional Requirements (NFRs)

4.1 Stability & Fault Tolerance (The "Immortal Boot" Guarantee)

- **Zero-Unbootable Policy:** System updates must automatically create a Btrfs pre-transaction snapshot. If an update breaks the kernel or graphics driver, the bootloader (GRUB) must expose an instant rollback menu allowing full recovery into the pre-update state in < 10 seconds.
- **Safe Driver Recompile:** NVIDIA driver DKMS compilation must execute during package installation. If the driver compilation fails, the transaction must halt before replacing the running kernel.

4.2 Performance & Responsiveness

- **Cold Boot Time:** 8 seconds from UEFI bootloader to SDDM login screen on standard NVMe PCIe Gen3+ SSDs.
- **Idle Memory Footprint:** 950 MB RAM consumption at cold boot with KDE Plasma running on Wayland.

- **Kernel Scheduling Latency:** Kernel tuned with `CONFIG_PREEMPT` / Zen scheduling patches to maintain consistent frame pacing under heavy background CPU load.
- **Memory Pressure Resilience:** Dynamic ZRAM enabled with `zstd` compression algorithm allocating 50% of physical RAM as swap space to prevent Out-Of-Memory (OOM) lockups.

4.3 Hardware Compatibility

- **Display Configurations:** Independent fractional scaling per monitor on Wayland without blur on legacy X11 applications.
- **Audio Routing:** PipeWire routing with automatic sample rate switching matching hardware capabilities without crackling or high latency.
- **Dual-Boot Interoperability:** Hardware RTC clock automatically configured to coordinate with Windows to eliminate the standard 5.5-hour time drift bug.

5. Security & Isolation Model

1. Rootless by Default:

- User account provisioned with password-less `sudo` restrictions.
- Container engines (Docker/Podman) configured for daemonless/rootless execution without requiring `sudo docker`.

2. Application Sandboxing:

- GUI applications distributed via Flathub run under Flatpak sandbox permissions managed through Plasma System Settings.

3. Immutable Boot Snapshots:

- Read-only boot snapshots mounted with copy-on-write integrity to prevent ransomware or rogue installer scripts from compromising recovery points.

6. Build & Delivery Architecture

- **Host-Agnostic Build Engine:** The ISO generation toolchain runs within an isolated container, enabling the OS to be built and verified on Linux servers, GitHub Actions CI/CD, or macOS developer workstations.
 - **Target Deliverable:** Bootable hybrid ISO (`doomos-plasma-x86_64.iso`) compatible with UEFI Secure Boot and legacy BIOS (CSM).
 - **Installer:** Fully customized **Calamares** graphical installer providing automated partitioning (Btrfs subvolumes layout `@`, `@home`, `@snapshots`, `@var_log`), timezone auto-detection, and driver selection.
-

7. Phased Release Roadmap

Phase 1: Foundation (M1)

- └─ Base Kernel + ZRAM + Btrfs Layout
- └─ KDE Plasma 6 Wayland Minimal Base
- └─ SDDM with DoomOS Theme & Autologin

Phase 2: The Core Problem Fixers (M2)

- └─ Snapper Automated Snapshot Hooks in GRUB
- └─ PipeWire Screen Sharing & Audio Loopback
- └─ Developer Sysctl tuning & Rootless Docker
- └─ NVIDIA DKMS Guardian Scripts

Phase 3: Gaming & Polishing (M3)

- └─ Wayland Explicit Sync & HDR Auto-calibrator
- └─ Game Controller (Xbox/PS5) Drivers + OpenRGB
- └─ Calamares Installer Customization & Btrfs Provisioning
- └─ Flathub Verified App Store Integration

Phase 4: ISO Mastering & Verification (M4)

- └─ Automated Build Pipeline (Dockerfile + archiso/live-build)
- └─ QEMU Virtual Machine Verification Test Suite
- └─ Golden Master ISO Release (v1.0.0)